

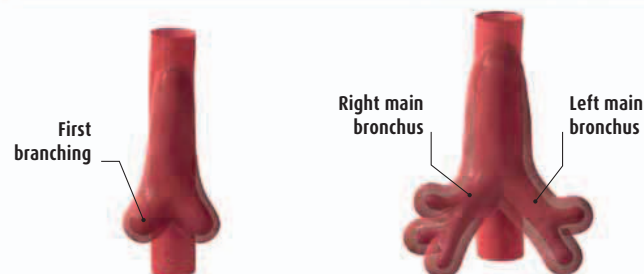
Q When are my baby's lungs fully formed? How can his respiratory system work when there's no air?

By about 36 weeks, your baby's lungs are developed enough for fully independent breathing (this is why babies born from 37 weeks onward are considered full term), but his lungs can't inflate until there is contact with air, so "breathing" doesn't start until after birth.

Our respiratory system consists not only of our lungs, but also the nose, pharynx, larynx, and trachea. The whole respiratory tract begins life as a long tube—in fact, it's the same tube that also gives rise to most of the digestive system (see p.157). By 36 weeks, most babies' lungs are fully formed for birth, and they contain plenty of surfactant, a substance that will allow the lungs to expand and take in air without collapsing, which makes independent breathing possible. Premature babies are supported by artificial breathing apparatus, such as a ventilator, or their oxygen levels will be controlled in an incubator.

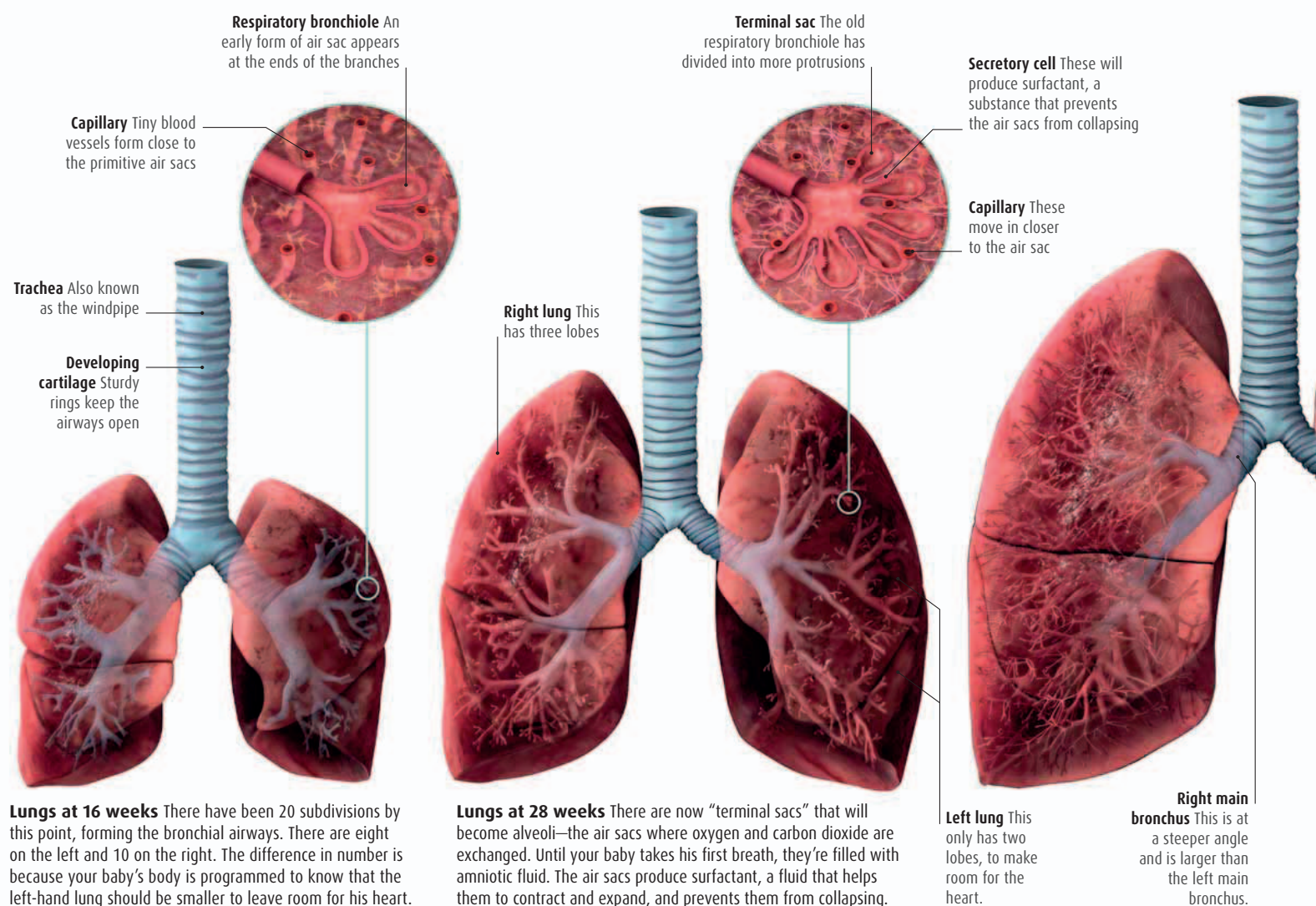
The diagrams show how the lungs start as buds that gradually branch out and complete their development at around 36 weeks of pregnancy.

FORMATION OF THE LUNGS



Lungs at 5 weeks The lung bud branches to form left and right main bronchi. Each of the bronchial buds will develop into a lung.

Lungs at 6 weeks The two bronchi develop differently: the left bronchus branches into two buds, and the right bronchus branches into three.



Lungs at 16 weeks There have been 20 subdivisions by this point, forming the bronchial airways. There are eight on the left and 10 on the right. The difference in number is because your baby's body is programmed to know that the left-hand lung should be smaller to leave room for his heart.

Lungs at 28 weeks There are now "terminal sacs" that will become alveoli—the air sacs where oxygen and carbon dioxide are exchanged. Until your baby takes his first breath, they're filled with amniotic fluid. The air sacs produce surfactant, a fluid that helps them to contract and expand, and prevents them from collapsing.

Left lung This only has two lobes, to make room for the heart.

Right main bronchus This is at a steeper angle and is larger than the left main bronchus.